

DA4 Term Project: Manufacturing Value Added and Water Scarcity

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Data Analysis 4

March 1, 2025

**To What Extent Does Manufacturing Value Added Contribute
to Water Scarcity?**

1. Introduction

This study investigates the extent to which manufacturing value added (MVA) contributes to water scarcity. Understanding this relationship is crucial, as industrial activities can significantly impact water resources, influencing both environmental sustainability and economic development. By analyzing this causal link, policymakers can better balance industrial growth with the preservation of vital water resources. For instance, rapid industrialization in water-scarce regions like northern China and parts of India has heightened concerns over sustainable water management. This study aims to quantify the extent to which manufacturing growth exacerbates water stress. This study hypothesizes that higher MVA contributes to increased water stress but that its impact may be attenuated when controlling for infrastructure (basic water services) and environmental shocks (natural disasters).

2. Data Description and Sources

The analysis utilizes data from several reputable sources:

Table 1: Summary of Variables and Descriptive Statistics

Variable	Description	Mean	Std Dev	Min	Max
Key Variables					
Year	Year of observation	2010.50	6.52	1999	2021
Country	Number of countries	-	-	-	-
Model Variables					
Water Stress (%)	Percentage of total renewable water resources used	72.77	309.34	0.03	3850.50
MVA (%)	Share of manufacturing in GDP	12.98	6.87	0.23	48.44
Disaster Count	Annual count of natural disasters	1.50	2.79	0	31
Water Stress (Box-Cox)	Transformed water stress variable	1.98	0.93	0.03	4.38

- **Manufacturing Value Added (MVA):** Data on the share of manufacturing in GDP is sourced from Our World in Data, which compiles information from the World Bank and OECD.
- **Water Stress:** Water scarcity data is extracted from the Food and Agriculture Organization’s AQUASTAT database.
- **Natural Disaster Counts:** The count of natural disasters per country-year is obtained from the Emergency Events Database (EM-DAT).
- **Access to Basic Water Services:** Data on the percentage of the population using at least basic drinking water services is retrieved from the World Bank’s World Development Indicators.

The dataset covers the period from 1999 to 2021, with annual observations for most variables. Missing values are imputed through interpolation. The Box-Cox transformation was used over winsorizing to improve normality and variance stability.

3. Empirical Models

To assess the causal impact of MVA on water stress, the following panel data models are employed:

1. Difference-in-Differences (DiD) Models

The following models estimate the impact of MVA on water stress using a DiD framework, where 2015 is considered the post-treatment period, and treatment is defined as countries above the median MVA value.

- **Basic DiD Model:** Estimates the direct effect of treatment and time on water stress:

$$\text{WaterStress}_{it} = \beta_0 + \beta_1 \text{Treatment}_{it} + \beta_2 \text{Post}_t + \beta_3 (\text{Treatment}_{it} \times \text{Post}_t) + \epsilon_{it} \quad (1)$$

- **DiD with Regional Fixed Effects:** Includes regional fixed effects:

$$\text{WaterStress}_{it} = \beta_0 + \beta_1 \text{Treatment}_{it} + \beta_2 \text{Post}_t + \beta_3 (\text{Treatment}_{it} \times \text{Post}_t) + \sum_r \delta_r \text{Region}_r + \epsilon_{it} \quad (2)$$

- **DiD with Confounders:** Extends the previous model with key confounders:

$$\begin{aligned} \text{WaterStress}_{it} = & \beta_0 + \beta_1 \text{Treatment}_{it} + \beta_2 \text{Post}_t + \beta_3 (\text{Treatment}_{it} \times \text{Post}_t) \\ & + \beta_4 \text{BasicWaterServices}_{it} + \beta_5 \text{DisasterCount}_{it} + \sum_r \delta_r \text{Region}_r + \epsilon_{it} \end{aligned} \quad (3)$$

2. OLS with Regional Fixed Effects

$$\text{WaterStress}_{it} = \beta_0 + \beta_1 \text{MVA}_{it} + \beta_2 \text{DisasterCount}_{it} + \beta_3 \text{BasicWaterServices}_{it} + \sum_r \delta_r \text{Region}_r + \epsilon_{it} \quad (4)$$

3. Fixed Effects Panel Data Regressions

$$\text{WaterStress}_{it} = \beta_0 + \beta_1 \text{MVA}_{it} + \sum_t \delta_t \text{Year}_t + \sum_i \gamma_i \text{Country}_i + \epsilon_{it} \quad (5)$$

4. Lagged Models

- **Two-Year Lagged Model:**

$$\begin{aligned} \text{WaterStress}_{it} = & \beta_0 + \beta_1 \text{MVA}_{it} + \beta_2 \text{MVA}_{it-2} + \beta_3 \text{WaterStress}_{it-2} \\ & + \beta_4 \text{DisasterCount}_{it} + \beta_5 \text{BasicWaterServices}_{it} + \epsilon_{it} \end{aligned} \quad (6)$$

- **Five-Year Lagged Model:**

$$\begin{aligned} \text{WaterStress}_{it} = & \beta_0 + \beta_1 \text{MVA}_{it} + \beta_2 \text{MVA}_{it-5} + \beta_3 \text{WaterStress}_{it-5} \\ & + \beta_4 \text{DisasterCount}_{it} + \beta_5 \text{BasicWaterServices}_{it} + \epsilon_{it} \end{aligned} \quad (7)$$

Table 2: Difference-in-Differences Regression Results

	Basic DiD		DiD + Regional FE		DiD + Confounders	
	(1)		(2)		(3)	
Intercept	9.477	***	6.660	***	2.030	(n.s.)
	(0.434)		(0.815)		(1.286)	
Treatment	1.268	***	1.246	(n.s.)	1.147	(n.s.)
	(0.053)		(0.176)		(0.121)	
Post	1.062	(n.s.)	1.075	(n.s.)	1.015	(n.s.)
	(0.059)		(0.073)		(0.081)	
Treatment:Post	0.911	(n.s.)	0.912	(n.s.)	0.947	(n.s.)
	(0.066)		(0.105)		(0.111)	
Basic Water Services					1.009	(n.s.)
					(0.007)	
Disaster Count					1.061	***
					(0.012)	
R²	0.013		0.387		0.421	
Observations	3051		3051		3051	

4. Results

Discussions

Table 2 presents the results of the Difference-in-Differences (DiD) regression models analyzing the impact of Manufacturing Value Added (MVA) on water stress. The basic DiD model (Column 1) suggests a statistically significant and positive association between the treatment group and water stress ($\beta = 1.268, p < 0.01$), indicating that areas with higher MVA exhibit increased water stress. However, after controlling for regional fixed effects (Column 2) and additional confounders (Column 3), this effect loses statistical significance, suggesting that other factors may be driving the observed correlation. The coefficient on the interaction term (*Treatment:Post*) remains positive but not statistically significant across all specifications. This suggests that the change in water stress following treatment is not robustly different from pre-treatment levels. Further, highlighting there are other drivers of water stress, such as other sectors withdrawals, environmental conditions and prior water stress levels.

The inclusion of disaster count as a control variable in Column 3 reveals a highly significant positive relationship ($\beta = 1.061, p < 0.01$), implying that natural disasters exacerbate water stress. Meanwhile, basic water services do not show a statistically significant effect, suggesting limited direct influence in this setting.

Manufacturing Value Added and Water Stress

Table 3 extends the analysis by estimating the direct impact of MVA on water stress using Ordinary Least Squares (OLS), fixed effects (FE), and lagged models. The OLS model (Column 1) shows a statistically significant positive relationship between MVA and water stress ($\beta = 1.016, p < 0.01$), suggesting that higher MVA is associated with greater water stress.

However, once country and year fixed effects are included (Column 2), the effect of MVA is no longer statistically significant, indicating that unobserved country-specific factors may be confounding the relationship. This aligns with the DiD findings, reinforcing the idea that the initial correlation observed in OLS may not imply causation.

The lagged models (Columns 3 and 4) test whether past values of MVA and water stress influence current water stress. Interestingly, the coefficient on water stress lagged by two and five years remains highly significant ($\beta = 2.997, p < 0.01$), implying strong persistence in water

Table 3: Regression Results: MVA and Water Stress

	OLS (1)	Fixed Effect (2)	2-Year Lag (3)	5-Year Lag (4)
Intercept	1.812*** (0.229)	2.735*** (0.1)	1.042*** (0.006)	1.10*** (0.015)
MVA	1.016*** (0.002)	1.005 (0.004)	1.004 (0.001)	1.002 (0.001)
MVA (Lag 2Y)	- -	- -	1 (0.001)	0.998 (0.001)
Water Stress (Lag 2/5Y)	- -	- -	2.997*** (0.004)	2.993*** (0.011)
Disaster Count	1.056*** (0.005)	- -	1 (0.00)	1.001 (0.001)
Basic Water Services	1.008*** (0.001)	- (0.001)	0.999*** (0.0)	0.999*** (0.00)
Fixed Effects	Region	Year, Country	None	None
Observations	3051	3051	2759	2759
R²	0.429	0.988	0.995	0.986

stress levels over time. This suggests that current water scarcity may be heavily influenced by past conditions rather than immediate manufacturing activities.

Potential Biases and Limitations

Despite the use of DiD and fixed effects models, some limitations persist:

- **Endogeneity Concerns:** Manufacturing share may be correlated with unobserved factors that also influence water stress, such as industrial policies, climate conditions, or regulatory frameworks.
- **Measurement Issues:** Water stress is influenced by multiple factors beyond MVA, including agricultural water withdrawals and infrastructure quality, which may not be fully accounted for in the model.
- **Short-Term vs. Long-Term Effects:** The lack of significance in the DiD models suggests that manufacturing's effect on water stress may not manifest immediately but rather over a longer horizon, as indicated by the significant lagged water stress coefficients.

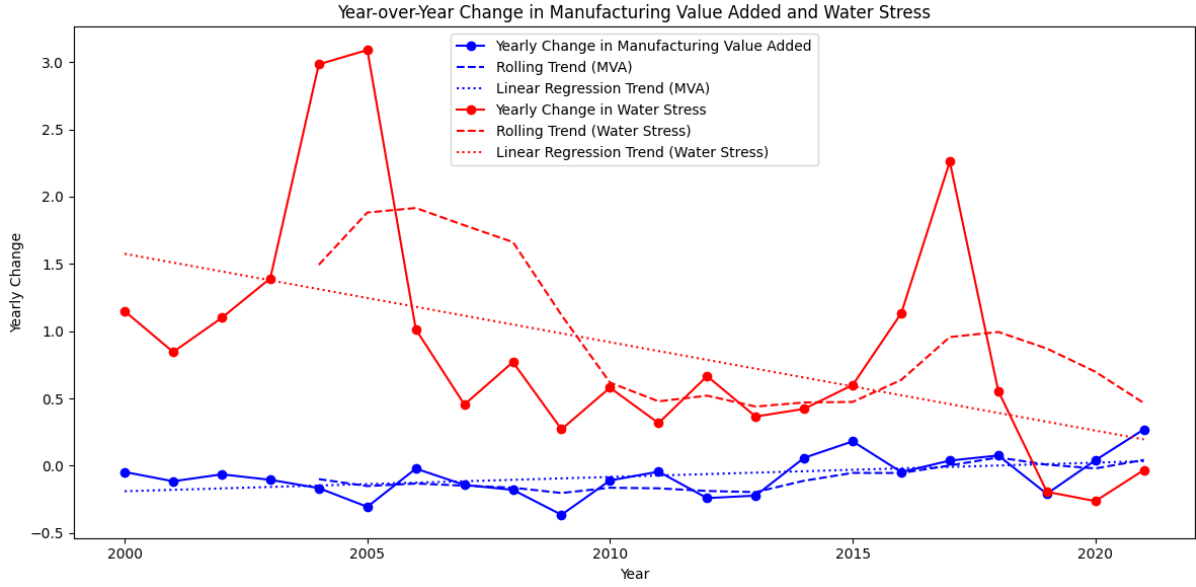
Another potential source of bias is reverse causality, regions with severe water stress might deter manufacturing investment, leading to an underestimation of MVA's impact on water stress.

Robustness Checks

The robustness checks incorporating fixed effects and lagged variables largely confirm the main findings:

- **Fixed Effects Models:** Controlling for unobserved heterogeneity removes the statistical significance of the MVA coefficient, suggesting that initial correlations in OLS were driven by country-specific factors.
- **Lagged Models:** The strong significance of past water stress levels reinforces the argument that water scarcity is path-dependent, meaning that past scarcity conditions strongly predict current stress levels.

Figure 1: Yearly Change in Manufacturing Value Added and Water Stress



4. Policy Implications and Conclusions:

The findings suggest that while manufacturing is correlated with water stress, its causal impact is less clear once confounding factors are accounted for. The evidence points to water stress being a persistent issue influenced more by past scarcity conditions and natural disasters rather than short-term changes in MVA. Given the strong persistence of water stress over time, policymakers should focus on long-term water conservation strategies rather than short-term interventions targeting manufacturing. The significant impact of natural disasters suggests that investments in climate resilience, such as improved water storage and emergency response systems, could help alleviate stress. Since industrial water use is only one component of overall water stress, policies should consider differential regulations across agriculture, domestic, and industrial sectors. While manufacturing's direct impact on water stress is inconclusive, encouraging water-efficient industrial processes and recycling technologies can help mitigate potential risks. Despite the unclear causal link, measures to be considered include: water pricing reforms, tax incentives for water efficient technology adoption, and public-private partnerships to improve water infrastructure. Overall, the evidence indicates that water scarcity is highly persistent and influenced more by historical conditions and extreme environmental events. These findings highlight the need for proactive, long-term water management policies rather than reactive strategies targeting short-term industrial activity.

A Appendix: Additional Figures and Analysis

To visualise the trend variations in water stress and MVA, in comparison to predicted values in the Fixed Effects model, please confer to the Jupyter Notebook.

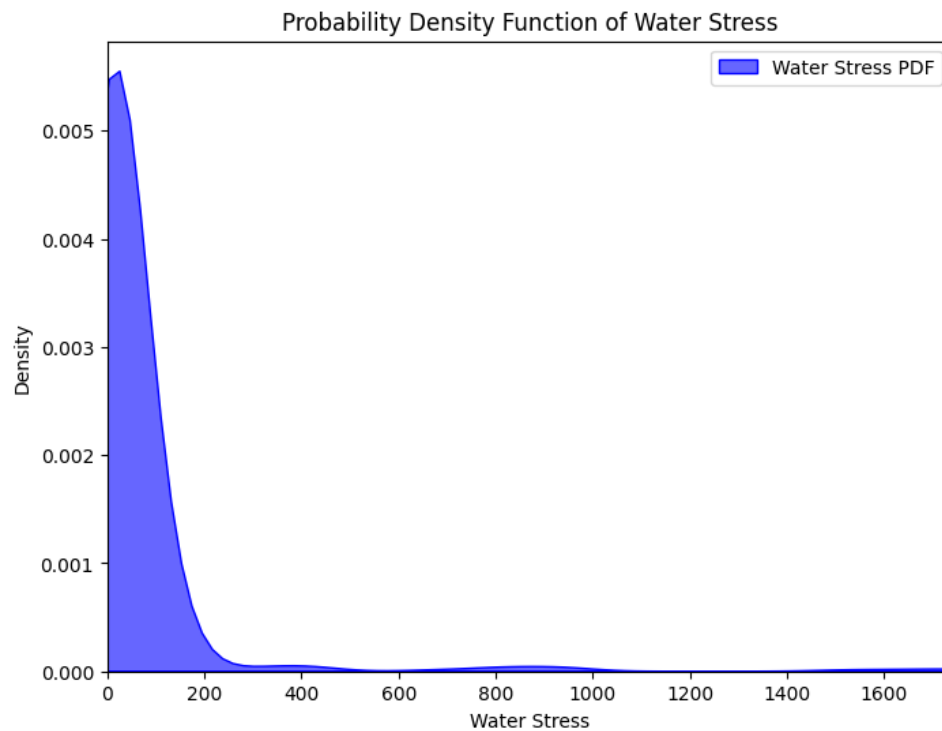


Figure 2: Probability Density Function (PDF) of Water Stress. This figure shows the distribution of water stress values.

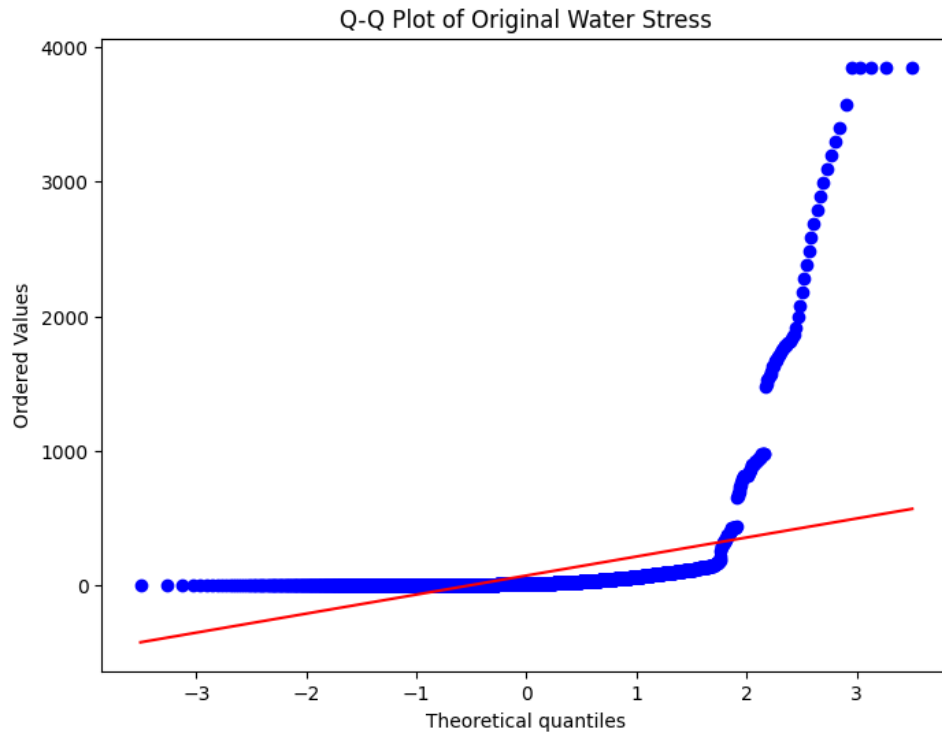


Figure 3: Q-Q Plot of Original Water Stress. This plot evaluates the normality of water stress data.

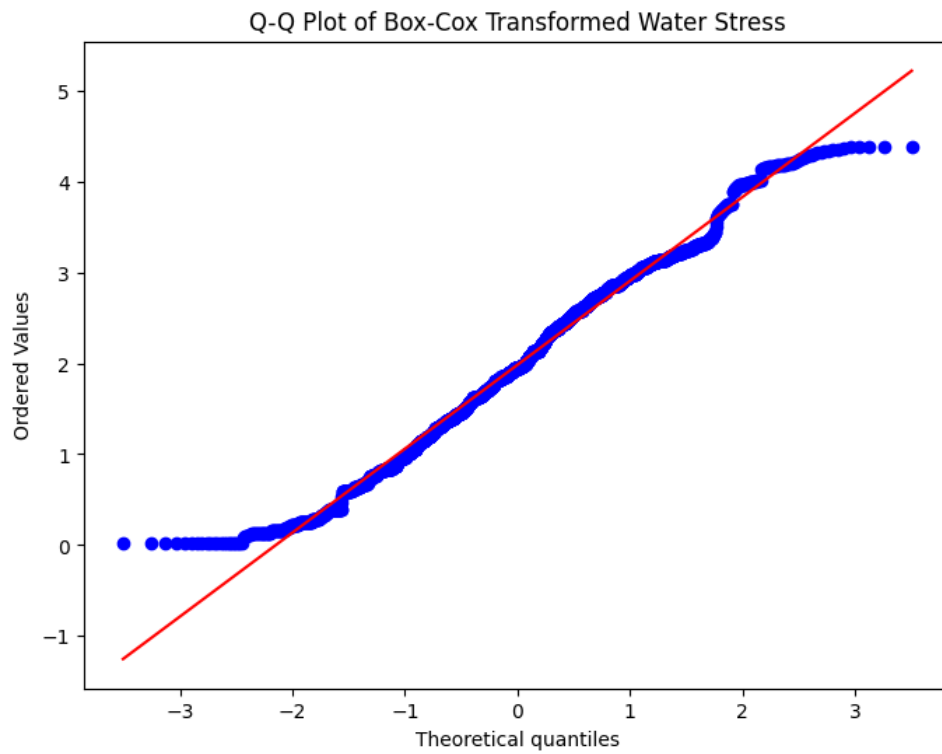


Figure 4: Q-Q Plot of Box-Cox Transformed Water Stress. This transformation improves normality.

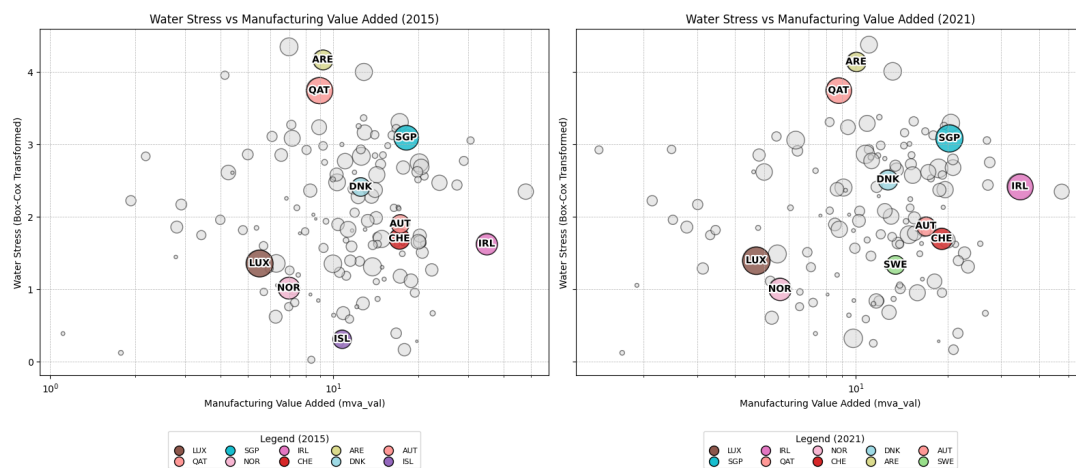


Figure 5: Scatterplot of Water Stress vs. Manufacturing Value Added (2015 & 2021). This visualization examines the relationship between MVA and water stress across countries.

Year-over-Year Changes and Trends for Water Stress, GDP, and Confounders

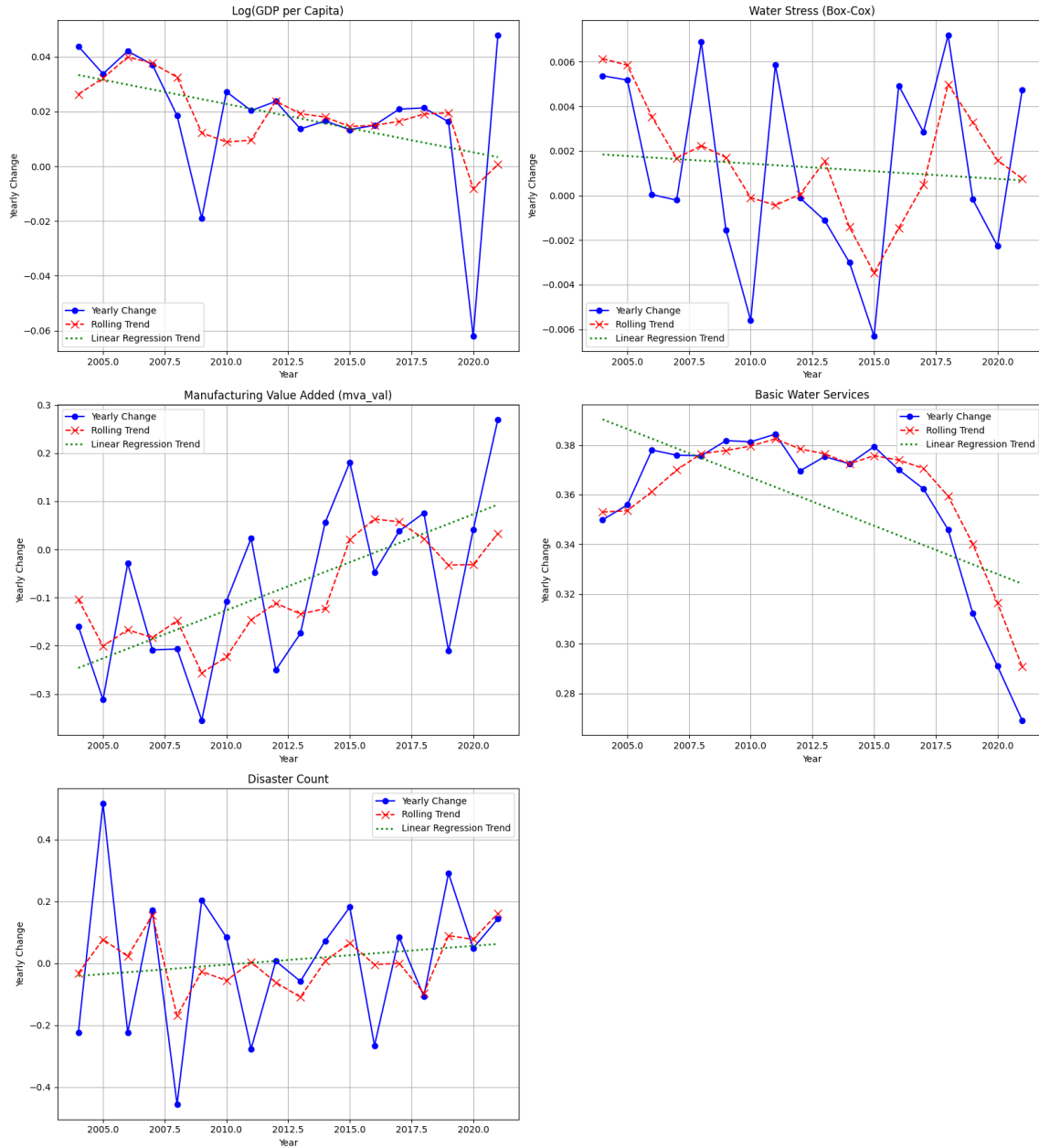


Figure 6: Year-over-Year Changes and Trends for Water Stress, GDP, and Confounders. This figure displays trends in key variables used in the analysis.

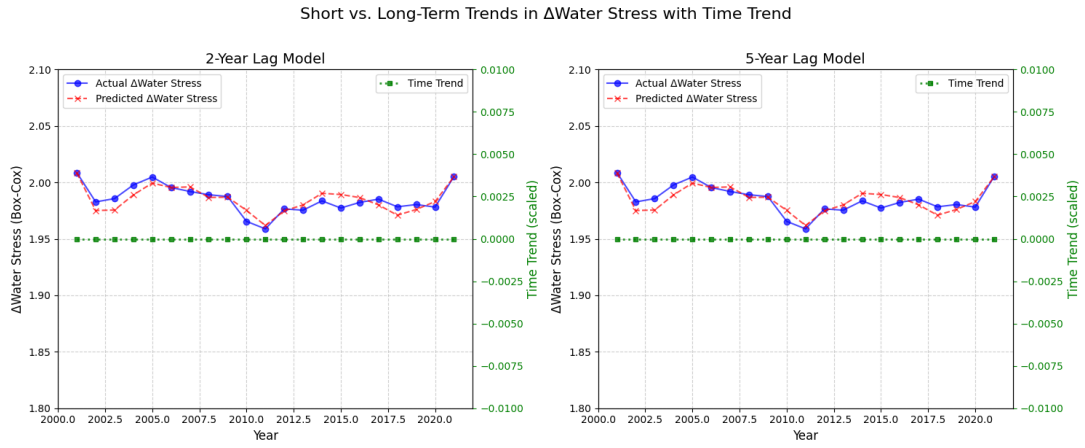


Figure 7: Short vs. Long-Term Trends in Water Stress with Time Trend. The plots show how water stress evolves over time using 2-year and 5-year lag models.